INDIAN INSTITUTE OF TECHNOLOGY DELHI
INTRODUCTION TO ELECTRICAL ENGINEERING
| AELL101 QUIZ-2 |

NAME: **ENTRY NO:**
DATE: 24-03-2025 (10 AM – 10:45 AM) **DURATION: 45 min**
Total marks: 15 **SET: A**

- ✓ **Each question carries equal marks. Space for rough work is given on the last page.**
- ✓ **Ask for more blank sheets, if necessary.**
- ✓ **MCQ (single choice): CIRCLE correct option, otherwise no marks.**

1. Three identical impedances are connected in delta. The load is supplied by a 3-phase supply connected in star with a phase-to-neutral voltage of 300V. One of the line currents is $30\sqrt{3}\angle 0$ A. Calculate the impedance per phase.

   A. 30 Ω B. 10 Ω
   C. $10\sqrt{3}$ Ω D. 20 Ω

2. Which of the following statements are TRUE?
   P: In a single-phase AC circuit, the total instantaneous power consumed by an RLC load is oscillating
   Q: In a balanced three-phase AC circuit, the total instantaneous power consumed by an RLC load is oscillating in nature.

   A. Only P B. Only Q
   C. Both P and Q D. None of these

3. The voltage and current expressions have been given below. Determine the power factor.

$$V(t) = 10cos(4t - 60°)\ v$$
$$I(t) = 4sin(4t + 50°)\ A$$

   A. 0.94 lagging B. 0.94 leading
   C. 0.985 lagging D. 0.985 leading

4. Which of the following statements are TRUE?
   P: Power is transmitted at high voltages to reduce I^2R losses
   Q: Power is transmitted at low voltages to reduce I^2R losses
   R: DC power system is preferred over AC because voltage transformation is possible in DC
   S: AC power system is preferred over DC because voltage transformation is possible in AC

   A. P and R B. P and S
   C. Q and R D. Q and S

5. Which of the following statements are TRUE about AC power?
   P: In a balanced 3-phase AC Y system, current in neutral wire is zero
   Q: A balanced AC 3-phase system can carry more power compared to DC system using the same amount of wires.
   R: DC power system is preferred over AC because voltage transformation is possible in DC
   S: AC power system is preferred over DC because voltage transformation is possible in AC

   A. P and R B. P and S
   C. Q and R D. Q and S

6. Identify TRUE statement for a star connected 3-phase AC circuit (only one).

   A. Phase voltage is equal to line voltage and phase current leads line current by 30^0

   B. Phase voltage leads line voltage by 30^0 and phase current is equal to line current

   C. Phase voltage lags line voltage by 30^0 and phase current is equal to line current

   D. Phase voltage is equal to line voltage and phase current lags line current by 30^0

7. Which of the following statements are TRUE?

   P: In case of capacitive circuit, capacitive reactance (Xc) is directly proportional to frequency

   Q: In case of capacitive circuit, capacitive reactance (Xc) is inversely proportional to frequency

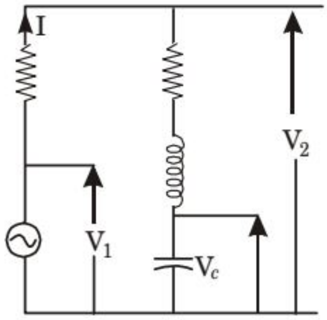
   R: In case of inductive circuit, inductive reactance (X_L) is directly proportional to frequency

   S: In case of inductive circuit, inductive reactance (X_L) is inversely proportional to frequency

   A. P and R　　B. P and S

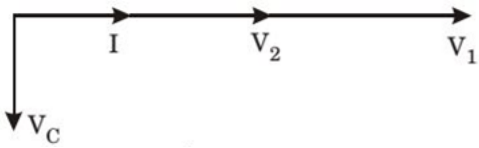
   C. Q and R　　D. Q and S

8. The circuit shown in the Figure below is energized by a sinusoidal voltage source V_1 at a frequency which causes $X_L = X_C$ with a current of I.

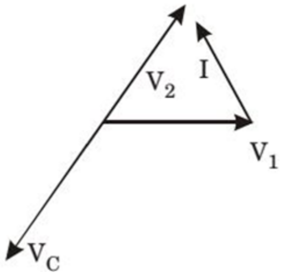


Which of the following phasor diagram is applicable to this circuit?

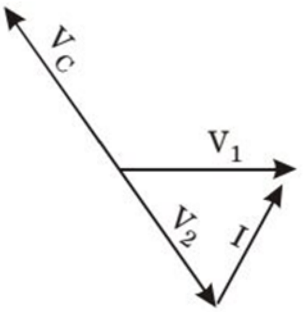
A.)



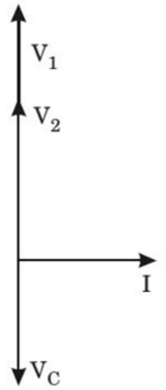
B.)



C.)

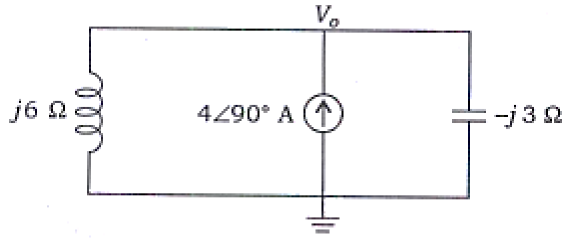


D.)



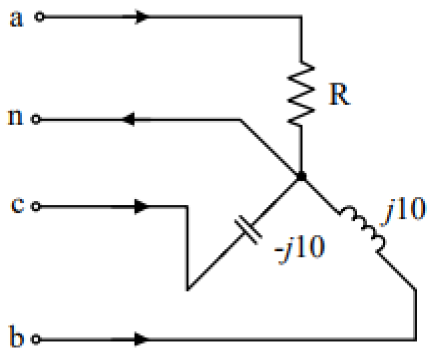
Numerical Type: Write only FINAL answer in the given field. Otherwise, no marks.

9. What will be the value of V_0 in the circuit shown in the following figure?



Ans:________________

10. A three-phase load is connected to a three-phase balanced supply as shown in the figure below. If Van = $100\angle 0°$ V, Vbn =$100\angle -120°$ V and Vcn =$100\angle -240°$ V then the value of R for zero current in the neutral wire is ___________Ω (up to 2 decimal places)



Ans:________________

11. A series RLC resonant circuit has R = 2 kΩ, L = 40 mH, and C = 1 μF. Find the Quality Factor.

Ans:________________

12. A parallel RLC resonant circuit has R = 5kΩ, L = 8 mH, and C = 60 μF. If the magnitude of applied rms voltage to the circuit is 25 kV then the magnitude of current (rms) flowing in inductor will be__

Ans:________________

13. What will be the total active power consumed by a 3-phase, delta-connected load, which is supplied with a line voltage of 230 V, when the value of the phase current is 15A and the current lags the voltage by 30°?

Ans: _________

14. A generator is supplying a load of 200 kW at 0.5 lagging power factor. If power factor is raised to unity, how many more kW can the generator supply assuming the kVA rating remains same.

Ans: _________

15. What would be the rms value of the following voltage signal: V = 50 sin(377t+30) ?

Ans: _________

Rough work space: